

PAWAN ALLEM

Applied AI Engineer • Agentic AI Systems Engineer • LLM Engineer

Hyderabad, India | +91-8501819565 | allem.pawan@gmail.com

[linkedin.com/in/pawan-allem](https://www.linkedin.com/in/pawan-allem) | pawanallem.netlify.app | github.com/AllemPawan

EXECUTIVE SUMMARY

Applied AI Engineer with 4+ years building production AI systems, now focused on Agentic AI engineering — local-first platforms, intelligent developer tools, and autonomous workflows using LangGraph, LangChain, MCP, Ollama, FastAPI, and modern LLMs.

Currently architecting an Internal AI Productivity Platform adopted by a 10-member engineering team, cutting time spent on documentation, code understanding, and reporting by 75%.

Focused on building AI systems that reason, retrieve, plan, and execute real work — not isolated demos.

SELECTED IMPACT

- 4+ years delivering production AI solutions for enterprise and government organizations.
- Architected a local-first AI platform that cut time on documentation, code understanding, and reporting by 75% for a 10-member engineering team.
- Combined a strong software engineering foundation with modern Agentic AI and LLM orchestration to ship production systems, not prototypes.
- Built reusable AI automation capabilities — from research agents to RAG platforms — now embedded in daily engineering workflows.

CORE EXPERTISE

Agentic AI	LangGraph, LangChain, AI Workflow Automation, MCP, Tool Calling, Deep Research Agents, Prompt Engineering
LLM Engineering	Gemma 4, Qwen Coder, Ollama, Local LLMs, RAG, Embeddings, Structured Outputs
AI Engineering	Python, FastAPI, REST APIs, SQL, Docker, Git, Modular Architecture
Machine Learning	PyTorch, TensorFlow, TensorFlow Lite, Computer Vision, OCR, YOLO, Scikit-learn
Focus Areas	Developer Productivity, Autonomous Research Systems, AI Automation, Intelligent Workflows

PROFESSIONAL EXPERIENCE

Software Development Engineer (Applied AI) | Vassar Labs

May 2022 – Present

- Architected and delivered production AI systems spanning intelligent automation, computer vision, edge AI, and enterprise software.
- Designed and built an Internal AI Productivity Platform (LangGraph, LangChain, Ollama, FastAPI, MCP, Gemma 4, Qwen Coder) that cut time on documentation, repo exploration, code explanation, and report generation by **75%** for a 10-member engineering team.
- Engineered a secure local-first AI architecture routing routine engineering work to local LLMs while reserving Claude Code for architecture, complex implementation, and higher-order reasoning — reducing dependency on external API costs.
- Partnered directly with engineering leadership to identify repetitive knowledge work and convert it into reusable AI capabilities adopted team-wide.
- Built modular AI infrastructure prioritizing extensibility and maintainability, enabling faster rollout of new AI capabilities across projects.

SELECTED AI SYSTEMS

- **AI Developer Copilot:** Internal AI assistant automating documentation, code understanding, file discovery, terminal commands, report generation, and reusable engineering workflows through a unified dashboard.
- **Autonomous Research Engineer:** CrewAI workflow that searches arXiv, downloads papers, extracts methodology, and generates developer-ready implementation reports using a fully local Gemma 4 model.
- **Deep Research Agent:** LangGraph-based autonomous research system implementing planning, iterative reasoning, retrieval, and tool orchestration inspired by modern Deep Research architectures.
- **Enterprise Knowledge Intelligence Platform:** Production-ready RAG platform (LangChain, embeddings, vector retrieval, FastAPI) for citation-grounded enterprise knowledge discovery.

EDUCATION, AWARDS & PROFESSIONAL DEVELOPMENT

- B.Tech., Computer Science Engineering — GITAM University
- Oracle AI Foundations Associate
- freeCodeCamp Python Certification
- Best Employee Award — Vassar Labs (2022 & 2023)
- Continuous research in Agentic AI, LangGraph architectures, Deep Research systems, and AI engineering best practices.